Kinetic Inductance Detector Nonlinear Resonance Fitting

Copy from Logan Foote Thesis

Foote, Logan Michael. “High-Sensitivity Superconducting Detectors for Far-Infrared Space Astrophysics.” Version Final. California Institute of Technology, 2025. PDF. https://doi.org/10.7907/3TM8-QX47.

Appendix B

KINETIC INDUCTANCE DETECTOR RESONATORS

Figure 2.1 is a schematic of the Kinetic Inductance Detector (KID) resonant circuit. The resonator is represented by the impedance Z and the coupling capacitor is C_c. The resonator is typically an absorber (for which the impedance is derived in Section 2.4) in parallel with a capacitor. Sometimes, an additional capacitor to ground is included. The complex forward transmission S_{21} is measured from port 1 to port 2.

B.0.1 Basic Resonance Model

The basic formula for the transmission S_{21} through the resonant circuit is derived in many sources.[3,5,6] The transmission is

$$S_{21} = 1 - \frac{Q_r}{Q_c}\frac{1}{1 + 2jy}, \tag{B.1}$$

where Q_r is the total quality factor and Q_c is the coupling quality factor. The fractional spacing between the tone frequency f and the resonant frequency f_r is given by

$$x = \frac{f - f_r}{f_r}, \tag{B.2}$$

such that the number of line widths between the tone frequency and the resonant frequency is $y = Q_r x$. The total resonator quality factor is related to the coupling quality factor and the internal quality factor Q_i through

$$\frac{1}{Q_r} = \frac{1}{Q_c} + \frac{1}{Q_i}. \tag{B.3}$$

An example of the basic resonance model is plotted in Figure B.1.

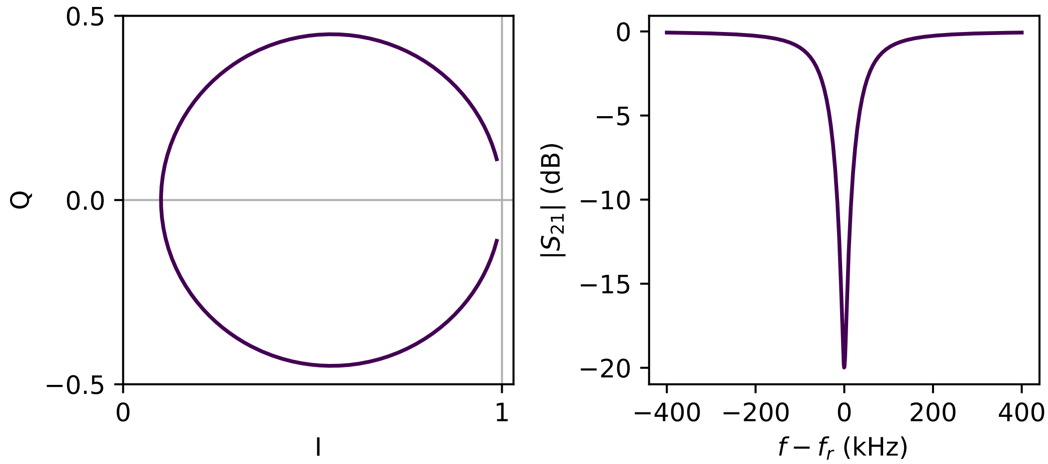

Figure B.1: Example of the basic resonance curve. (Left) imaginary (Q) versus real (I) component of the transmission. The resonance traces a circle with diameter Q_r/Q_c. (Right) transmission magnitude versus frequency. The resonance dip is centered on f_r and the FWHM is $\sqrt{3} f_r/Q_r$.

B.0.2 Impedance Mismatch

A mismatch in the impedance of the input and output lines can be modelled by introducing an imaginary component to the coupling quality factor through the angular parameter ϕ. The modification to the resonator transmission was calculated in Khalil et al. 2012.[4] The resonator transmission becomes

$$S_{21} = 1 - \frac{Q_r e^{j\phi}}{Q_c \cos\phi} \frac{1}{1 + 2jy}. \tag{B.4}$$

An example of the shift in the resonance caused by the impedance mismatch is plotted in Figure B.2.

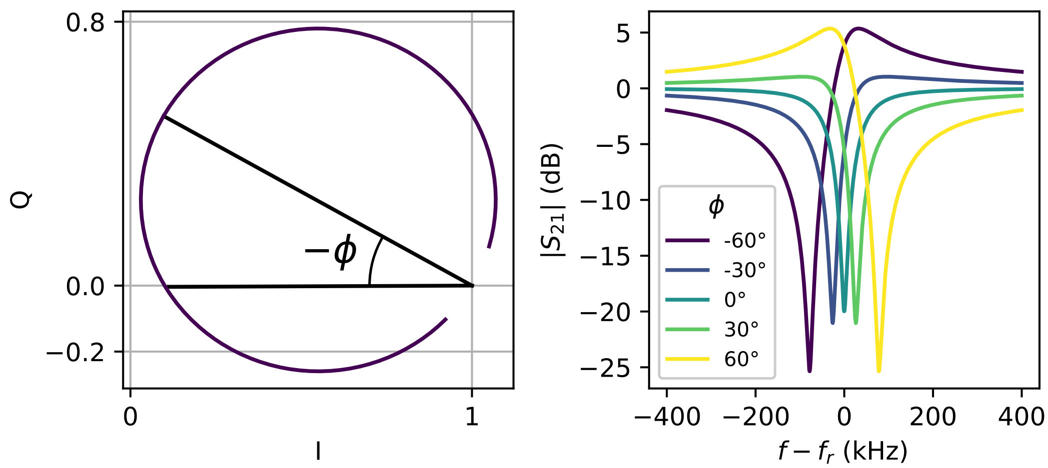

Figure B.2: Example of the shift in the resonance caused by an impedance mismatch. (Left) Q versus I. The angle ϕ determines the angle between the real axis and the location of the resonant frequency. (Right) transmission magnitude versus frequency for several values of ϕ.

B.0.3 Nonlinear Inductance

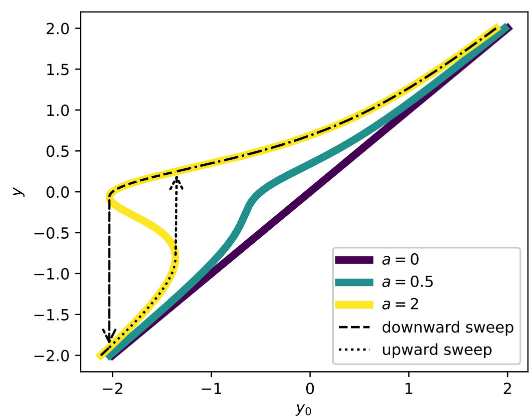

Figure B.3: y versus y_0, with paths drawn for upward and downward sweeps on the bifurcated curve.

The kinetic inductance is nonlinear, and microwave QP generation creates an additional nonlinearity. The nonlinear inductance can be modelled as the series

$$L(I) = L(I = 0)\left(1 + \frac{I^2}{I_2^2} + \cdots\right), \tag{B.5}$$

where odd powers of I vanish due to symmetry constraints and I_2 sets the scaling of the nonlinear effect. I_2 is on the order of the ratio of the inductive energy to the pairing energy.[8] The modification to S_{21} due to the nonlinear inductance is derived in Swenson et al. 2013.[7] The power in the resonator will change as a tone with constant power is swept in frequency. This behavior manifests as a modification to y in Equation B.4, where y becomes the solution to

$$y - \frac{a_{\mathrm{nl}}}{1+4y^2} = y_0 = Q_r \frac{f - f_{r,0}}{f_{r,0}}, \tag{B.6}$$

where the nonlinearity parameter a_{nl} is given by

$$a_{\mathrm{nl}} = \frac{2Q_r^3}{Q_c} \frac{P_{\mu\mathrm{W}}}{2\pi f_r E_\star}, \tag{B.7}$$

where $P_{\mu\mathrm{W}}$ is the microwave power and $E_\star \propto L_k I_2^2/\alpha^2$, where $\alpha = L_k/L_{\mathrm{total}}$ is the kinetic inductance fraction. This equation can be solved for y, and it reaches a critical value at $a_{\mathrm{nl}}^{\mathrm{bif}} = 4\sqrt{3}/9 \approx 0.77$. For $a_{\mathrm{nl}} \leq a_{\mathrm{nl}}^{\mathrm{bif}}$, only one real solution exists for all values of y_0. For $a_{\mathrm{nl}} > a_{\mathrm{nl}}^{\mathrm{bif}}$, a range of y_0 values produce three real solutions. Two of these three solutions are stable (the largest and smallest stored energy states), so the resonator is said to have undergone bifurcation. Bifurcation is a hysteretic effect: as the probe tone is swept from low to high frequency the resonance is pulled towards the tone until it reaches the single-solution regime and quickly jumps to the left, or as the probe tone is swept from right to left the resonance is pushed away from the tone until they overlap and the resonance quickly jumps to the right. Figure B.3 is a plot of the value of y versus the value of y_0. y can be thought of as the spacing between the tone frequency and the actual resonant frequency, in number of line widths. Similarly, y_0 can be thought of as the spacing between the tone frequency and the resonant frequency with $P_{\mu\mathrm{W}} = 0$, in number of line widths.

This nonlinear behavior makes KID readout impractical above bifurcation. In practice, KIDs are typically read out around $a_{nl} = 0.5$, where the power is as high as reasonably possible to improve clearance above amplifier noise without reaching bifurcation. An example of the effect of the nonlinear kinetic inductance on the resonance shape is plotted in Figure B.4.

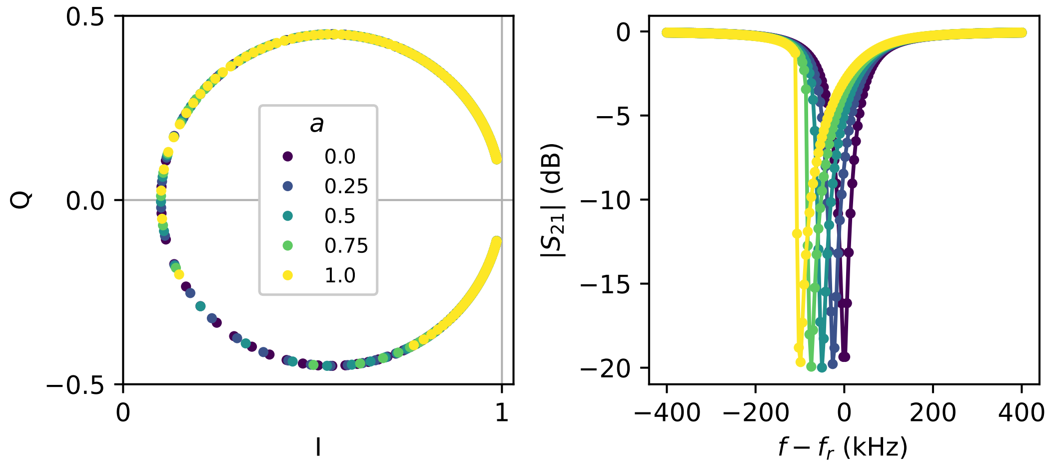

Figure B.4: Effect of the nonlinear kinetic inductance on resonance shape for a range of values of the nonlinearity parameter a_{nl}. (Left) Q versus I. (Right) transmission magnitude versus frequency.

B.0.4 Readout Circuit

The readout circuit gain can be written as

$$S_{21}^{\mathrm{readout}} = A(f)e^{-j2\pi f\tau_{\mathrm{cable}}+j\delta_0}, \tag{B.8}$$

where $A(f)$ is the frequency-dependent gain amplitude, τ_{cable} is the cable delay, and δ_0 is the phase offset.

B.1 Fitting

The resonance fitting routine is available in the `citkid` Python package,[2] which uses the Trust Region Reflective algorithm for fitting. Resonance data can be fit to the following equation:

$$S_{21} = A(f)e^{-j2\pi f\tau+j\delta}\left[1 - \frac{1}{1+2jy}\frac{Q_r}{Q_c\cos\phi}e^{j\phi}\right]. \tag{B.9}$$

In practice, the gain amplitude and phase are first removed from the data through polynomial fits to the data on either side of the resonance. A second-order polynomial fit is used for the gain amplitude, and a first-order fit is used for the gain phase (the cable delay and phase offset). After removing the gain amplitude and phase, the multiplicative complex parameter $z_0 = i_0 + jq_0$ and cable delay are kept in the fitting equation for fine-tuning. The fitting equation is

$$\boxed{S_{21} = (i_0 + jq_0)e^{-j2\pi f\tau}\left[1 - \frac{1}{1+2jy}\frac{Q_r}{Q_c \cos\phi}e^{j\phi}\right],} \tag{B.10}$$

where the eight parameters to fit are

$$\begin{aligned}
f_r &: \text{resonant frequency}\\
Q_r &: \text{resonant quality factor}\\
A &: Q_r/Q_c\\
\phi &: \text{impedance mismatch angle}\\
a_{\mathrm{nl}} &: \text{nonlinearity parameter}\\
i_0 &: \text{real component of gain}\\
q_0 &: \text{imaginary component of gain}\\
\tau &: \text{residual cable delay.}
\end{aligned}$$

B.1.1 Guessing Function

Parameter	Lower range	Upper range
f_r	10 MHz	10 GHz
Q_r	10^3	10^6
A	10^{-3}	$1 - 10^{-5}$
ϕ	$-\pi/2$	$\pi/2$
a_{nl}	0	1
$i_0,\ q_0$	-10	10

Table B.1: Simulated resonance data parameter ranges.

Robust resonance fitting requires a parameter initial guess function that performs well over a wide range of parameters. To assess the capability of the non-linear fitting algorithm, 100,000 model resonances were generated and fit. The ranges of the model parameters used in the simulation are given in Table B.1. Gaussian noise was added to each data point with randomized amplitudes for each resonance. The resulting simulated data sets are of the form $\{f, z\}$, where f is the frequency data in Hz and z is the complex S_{21} data. From the resulting data, the parameter guessing function was created. The following procedures are used to determine the initial guess.

1. z_0: the mean of the two lowest frequency and two highest frequency points in the sweep.

2. τ: 0. The cable delay should be close to 0 if the gain phase is removed properly.

3. ϕ: the IQ data is fit to a circle, and the angle of the vector from z_0 to the center of the circle is the guess for ϕ (see Figure B.2 (left)).

4. A: the guess for A is given by $A = 2R|\cos(\phi)|/|z_0|$, where R is the radius from the circle fit.

5. Q_r: First, ϕ and A are removed from the data through a rotation and the absolute value is taken to get

$$|z_{\text{rot}}| = \left|\left(1 - \frac{z}{z_0}\right) \frac{e^{-j\phi}\cos\phi}{A}\right|. \tag{B.11}$$

A Gaussian filter is applied to $|z_{\text{rot}}|$, and the full width at half maximum (FWHM) is extracted using a simple peak and width finding algorithm. The FWHM divided by the frequency of the peak is used to extract Q_r. A logarithmically-weighted linear fit to the logarithm of the Q_r and FWHM$/f_0$ data is performed to extract the Q_r guess, as plotted in Figure B.5 (left).

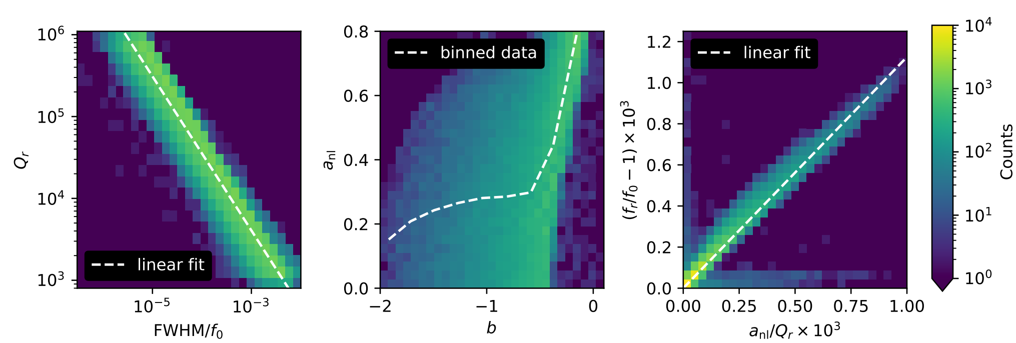

Figure B.5: (Left) Histogram of Q_r versus FWHM$/f_0$, where f_0 is the frequency at the peak. The linear fit is over-plotted. (Center): Histogram of a_{nl} versus b with binned data. (Right) Histogram of f_r versus a_{nl}/Q_r with linear fit.

6. a_{nl}: First, the data is rotated via Equation B.11. The quantity b is defined by

$$b = \left|\Delta z_{\text{rot}}^{\text{peak}}\right| \frac{f^{\text{peak}}}{\Delta f}, \tag{B.12}$$

where $\left|\Delta z_{\text{rot}}^{\text{peak}}\right|$ is the maximum spacing between adjacent points in IQ space, Δf is the frequency spacing (frequencies are assumed to be linearly spaced), and f^{peak} is the frequency corresponding to $\left|\Delta z_{\text{rot}}^{\text{peak}}\right|$. The resulting data is binned to create an array of data for interpolating from b to a_{nl}. a_{nl} versus b and the binned data are plotted in Figure B.5 (center). Resonators were generated up to $a_{\text{nl}} = 2$ and found to follow a near-vertical line with respect to b, so the upper cutoff of $a_{\text{nl}} = 1$ was chosen for the fitter.

7. f_r: f^{peak} is modified by a linear fit to f_r versus a/Q_r. The simulated data with a linear fit is plotted in Figure B.5 (right).

B.1.2 Bounds

The bounds for the fitting procedure are given in Table B.2.

Parameter	Lower bound	Upper bound
f_r	$\min f$	$\max f$
Q_r	$Q_r^{\text{guess}}/10$	$Q_r^{\text{guess}} \times 10$
A	10^{-3}	$1 - 10^{-6}$
ϕ	$-\pi/2$	$\pi/2$
a_{nl}	0	1
i_0	$i_0^{\text{guess}}/10$	$i_0^{\text{guess}} \times 10$
q_0	$q_0^{\text{guess}}/10$	$q_0^{\text{guess}} \times 10$
τ	-10^{-6}	10^{-6}

Table B.2: Bounds for the resonance fitting procedure, where f is the frequency data.

B.1.3 Evaluation of the Fit

The residuals from the fit to Equation B.10 are defined as

$$\text{res} = \frac{1}{N}\sqrt{\sum_{i}^{N} \left|z_i - z_i^{\text{fit}}\right|^2}, \tag{B.13}$$

where z_i is the data at index i, z_i^{fit} is the value of Equation B.10 at index i using the optimal fit parameters, and N is the total length of the data. Note that this definition of the residuals assumes that the gain is removed, and therefore (i_0, q_0) is close to

$(1, 0)$, such that the residuals of different datasets are comparable. Figure B.6 is a histogram of the residuals of fits to the simulated data. A threshold of 6×10^{-4} is defined to indicate poor fits.

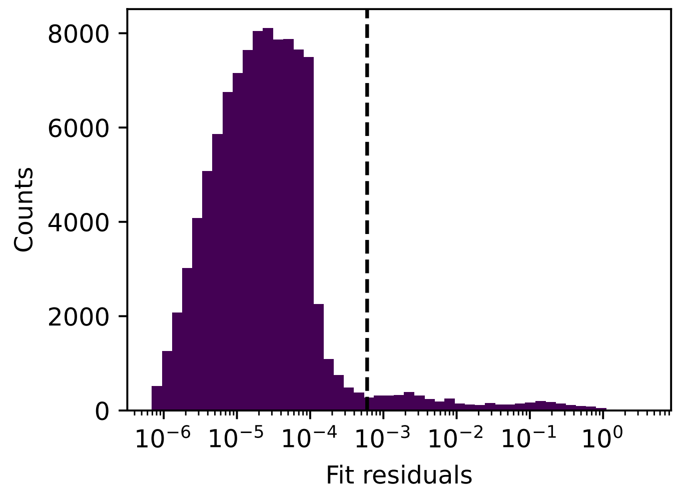

Figure B.6: Histogram of the residuals from fits to the simulated data. The median residual is 2×10^{-5}. The dashed black line indicates the threshold for poor fits.

A corner plot of the percentage of failed fits as a function of the five resonance fit parameters is plotted in Figure B.7. Overall, 95% of the fits were successful. Significant correlations between parameter values and failed fits are: $Q_r > 300,000$ (15% of failed fits), and $|\phi| > 0.9\pi/2$ (9% of failed fits). The remainder of the failed fits are distributed evenly throughout the parameter space.

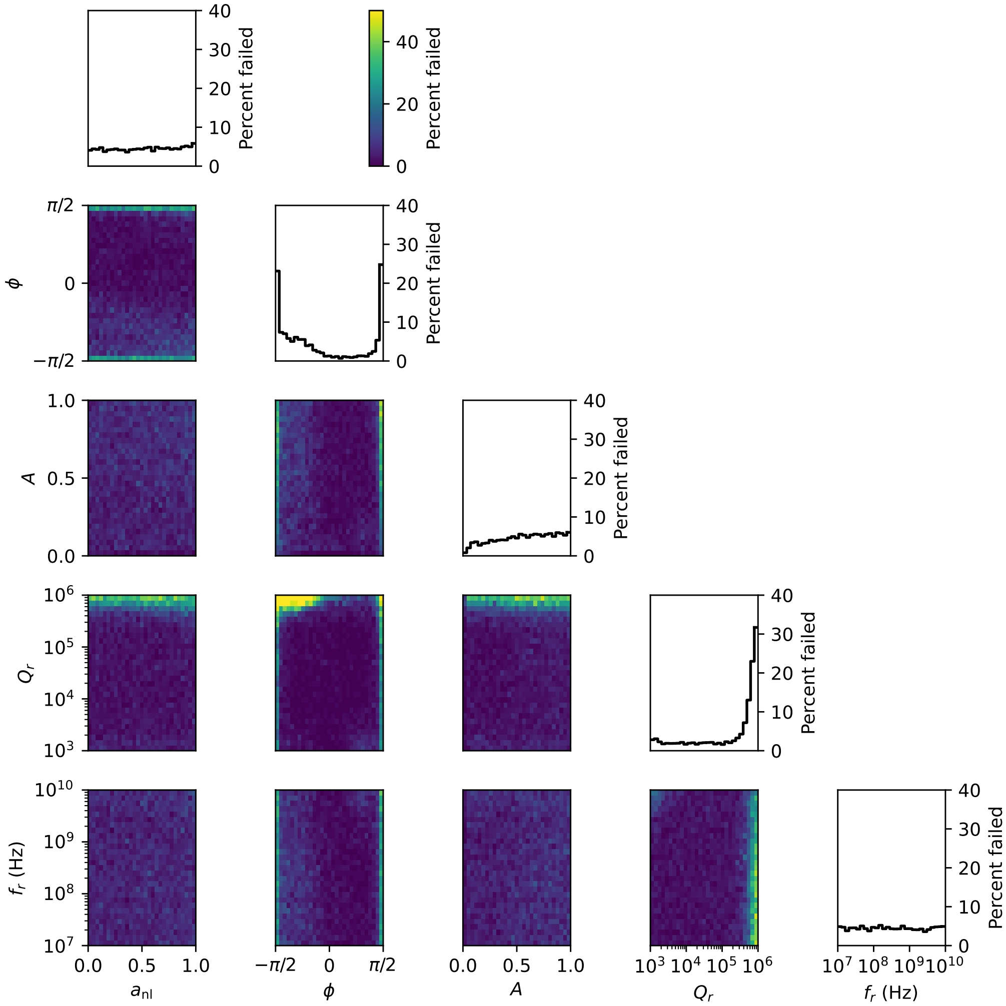

Figure B.7: Corner plot of failed fits versus parameter values. The histograms tally the percent of failed fits that fall into each bin.